

SIMILAR ANAEROBIC AND AEROBIC ADAPTATIONS AFTER 2 HIGH-INTENSITY INTERVAL TRAINING CONFIGURATIONS: 10 s:5 s vs. 20 s:10 s WORK-TO-REST RATIO

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ABSTRACT

Moghaddam, M, Estrada, CA, Muddle, TWD, Magrini, MA, Jenkins, NDM, and Jacobson, BH. Similar anaerobic and aerobic adaptations after 2 high-intensity interval training configurations: 10 s:5 s vs. 20 s:10 s work-to-rest ratio. *J Strength Cond Res* XX(X): 000–000, 2019—This study compares the effects of 2 high-intensity interval training (HIIT) configurations, a 10-5 vs. a 20-10 second work-to-rest ratio, on anaerobic and aerobic performance. Thirty-four individuals were randomly assigned to 10-5-HIIT ($n = 17$) and 20-10-HIIT ($n = 17$) groups to complete 6 cycles of 6 exercises, 3 days a week for 4 weeks. The 10-5-HIIT was performed with 10 s:5 s work-to-rest ratio with 1-minute recovery between cycles, while the 20-10-HIIT was performed with 20 s:10 s work-to-rest ratio followed by a 2-minute recovery. Anaerobic (i.e., peak power [PP], anaerobic capacity [AC], anaerobic power [AP], and total work [TW]) and aerobic fitness (i.e., time to exhaustion [TE], absolute $\dot{V}O_{2\max}$ [$A\text{-}\dot{V}O_{2\max}$], relative $\dot{V}O_{2\max}$ [$R\text{-}\dot{V}O_{2\max}$]) were measured with pre-training and post-training intervention. A significant main effect time was observed for both 10-5-HIIT and 20-10-HIIT ($p < 0.05$) in PP (9.2%, 5.7%); AC (14.9%, 8.6%); AP (9.0%, 6.2%); TW (15.1%, 8.5%); TE (4.3%, 5.5%); $A\text{-}\dot{V}O_{2\max}$ (9.4%, 8.9%); $R\text{-}\dot{V}O_{2\max}$ (8.5%, 8.2%), respectively. In conclusion, individuals may be able to achieve similar health benefits as 20-10-HIIT by performing 10-5-HIIT, despite exercising for 50% less total time. High-intensity interval training has been suggested as a “time-efficient” mode of exercise that can mitigate the most significant barrier to physical activity, “lack of time.” Both 10-5-HIIT and 20-10-HIIT can induce performance adaptations to a similar extent. However, because of shorter time commitment,

performing 10-5-HIIT at 10 s:5 s work-to-rest ratio may offer a shorter and equally efficient interval. Functional fitness training during HIIT protocols seems to be as beneficial as ergometer-based HIIT to improve anaerobic and aerobic performance.

KEY WORDS HIIT, muscular power, $\dot{V}O_{2\max}$, functional fitness exercise, Wingate

INTRODUCTION

The various elements of physical fitness, including body composition, flexibility, cardiorespiratory, muscular, and neuromotor fitness, have significant impact on physical health. The quality and frequency of physical activity have been identified as the main driver for the development and maintenance of each element of physical fitness in healthy adults (8). According to the American College of Sports Medicine (ACSM) (8), it is suggested that adults younger than 65 years must perform at least 30–60 minutes of moderate-intensity exercise (5 days a week at 64–76% of maximal heart rate [HR_{max}]), or 20–60 minutes of vigorous-intensity exercise (3 days a week at 77–95% of HR_{max}), to meet the recommended cardiorespiratory exercise frequency and duration. Furthermore, it is suggested that adults must engage in resistance, flexibility, and neuromotor training (known as “functional fitness training”) for at least 2 or 3 days per week (8). This is while the percentages of US adults (aged 18 years and older) who meet the recommended physical activity for aerobic exercise and combination of aerobic and resistance trainings are 51.7 and 21.7%, respectively (4). There are some personal and environmental barriers associated with physical inactivity such as lack of access to facilities, lack of social support, lack of motivation, as well as “lack of time,” which has been reported as the main barrier for meeting the recommended level of physical activity (20). Consequently, short-duration and low-volume exercise protocols have increasingly become popular for maintaining and improving physical fitness (27).

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High-intensity interval training (HIIT) has recently become popular among athletes and fitness professionals. According to the Worldwide Survey of Fitness Trends (24), HIIT is anticipated to be the most popular fitness trend for 2018, whereas it has been among top 3 fitness trends since 2014 (24). High-intensity interval training refers to a cycle of short bursts of “all-out” exercises separated by short rest periods. Each bout of exercise may last a few seconds up to several minutes, and completing a HIIT training session usually takes less than 30 minutes (1). The “time efficiency” due to lower exercise volume and duration as well as similar cardiorespiratory adaptations makes HIIT a great alternative for moderate-intensity continuous training (MICT) (1).

Most HIIT protocols typically include either running- or ergometer-based modes of exercise, which are performed near HRmax to elicit the desired physiological responses (2). However, the physiological adaptations of HIIT may vary due to factors such as (a) intensity of each bout; (b) duration of each bout; (c) active or inactive rest periods after each bout; (d) duration of rest periods after each bout; (e) number of bouts in each cycle; (f) number of cycles; (g) active or inactive recovery after each cycle; (h) duration of recovery after each cycle; and (i) exercise mode (2). From this list, the intensity and duration of both exercise and rest portions of each cycle are suggested as the most significant factors for optimizing the adaptations to HIIT (2). A significant challenge, despite these developments, is that different studies use different intensities and durations such as a variety of fixed work-to-rest ratios (i.e., 2:1, 1:1, 1:2), or a fixed percentage of HRmax (15), and thus, it is difficult to compare the adaptations to different HIIT protocols. The reason is that the physiological responses may vary due to the differences in the nature of the exercise regime. For example, a HIIT protocol with shorter work periods performed at supramaximal intensity may elicit greater adaptations specific to the ATP-PCr system compared with an HIIT protocol with longer work periods performed at a maximal intensity. Currently, no optimum HIIT protocol has been established to determine the most efficient intensity, exercise mode, exercise volume, or recovery duration. However, it has been reported that a 2:1 work-to-rest ratio causes a higher accumulated oxygen deficit than other ratios (i.e., 1:1, 1:2) (15). Regarding 2:1 work-to-rest ratio, previous literature has reported that the most effective and shortest duration HIIT interval uses 20 s:10 s work-to-rest ratio (i.e., 20-10-HIIT) (15). Tabata et al. (23) initially studied the impact of HIIT on enhancing the physical fitness of Olympic sprint skaters. The HIIT protocol they developed was conducted on a cycle ergometer and consisted of 8 bouts of 20 seconds at 170% of $\dot{V}O_{2\max}$, followed by 10-second rest periods (23).

Given that the intensity and duration of HIIT intervals significantly influence its physiological adaptations (2), this study incorporated a shorter protocol of 10 s:5 s work-to-rest ratio (i.e., 10-5-HIIT) to establish a shorter yet potentially effective interval duration. In the last few decades,

ergometer-based HIIT has been suggested as an efficient method to improve cardiovascular fitness in athletes and adults (1,21). This is while HIIT based on whole-body training has been recently shown as effective as ergometer-based (14,16,21). According to the recent study, however, both body mass-based and ergometer-based HIIT, using the same protocol, have shown the percentage of heart rate peak close to 90%, showing their intensity is equivalent and can be classified as vigorous training due to the ACSM guidelines (21). Furthermore, functional fitness training modality is more practical using inexpensive equipment and body mass exercises. In addition, functional fitness training can be easily performed by a wide range of populations, including those without access to gym facilities or equipment. Therefore, this study chose to apply functional fitness training during the HIIT protocols, recommended in ACSM guidelines (8), as an alternative for running- or ergometer-based training. It was hypothesized that the 10-5-HIIT

protocol at 10 s:5 s work-to-rest ratio induces similar performance adaptations as the 20-10-HIIT protocol at 20 s:10 s work-to-rest ratio. It was also hypothesized that performing functional fitness training during HIIT is beneficial to improve anaerobic and aerobic performance. The purpose of this study was therefore to compare the effects of 10-5-HIIT and 20-10-HIIT protocols using functional fitness training on anaerobic and aerobic performance.

METHODS

Experimental Approach to the Problem

This study used a randomized, between-subject, pre-post experimental design (25). A 4-week intervention was implemented to compare the effects of the HIIT protocols. The tests included anthropometric measurements (i.e., body mass and height), anaerobic (i.e., Wingate test), and aerobic performance testing (i.e., graded exercise test). Regarding the randomization protocol, we divided the participants into 2 groups based on sex (i.e., men = 16 and women = 18). Each group was then entered in a random drawing to be assigned to either the 10-5-HIIT (8 men and 9 women) or 20-10-HIIT (8 men and 9 women) group.

Subjects

Thirty-four recreationally active individuals, with an age range of 19 to 28, (mean $\pm$ SD, men: $N = 16$; age = 21.6 ± 3.57 years; height = 177.3 ± 6.54 cm; mass = 86.1 ± 16.34 kg; women: $N = 18$; age = 20.6 ± 0.86 years; height = 165.7 ± 5.22 cm; mass = 67.2 ± 9.18 kg) participated in this study (Table 1). The study was approved by and performed in accordance with the recommendations of the Institutional Review Board at Oklahoma State University for the protection of human subjects (IRB Application #: ED-16-163). Before the start of the investigation, the participants completed the IRB-approved informed consent form as well as an Exercise Preparticipation Health Screening Process and the Physical Activity Readiness Questionnaire (PAR-Q and

TABLE 1. Participant demographics.*†

Variable	10-5-HIIT	20-10-HIIT
Gender	8 males, 9 females	8 males, 9 females
Age (y)	21.7 ± 2.9	20.4 ± 2.2
Height (m)	173.0 ± 9.1	169.5 ± 7.2
Body mass (kg)	82.1 ± 10.2	70.7 ± 10.3

*HIIT = high-intensity interval training.

†Mean ± SD.

You) to determine their health status and ability to participate. Specifically, afterward, the participants were familiarized with the testing and training procedures and techniques.

Procedures

Anthropometric Measurements. The participants were asked to step barefoot on the weighing platform and stadiometer to measure their body masses and heights, respectively. The body masses were recorded in kilograms (kg), and the heights were measured in centimeters (cm).

Anaerobic Fitness. After the completion of the anthropometry measurements, each participant performed the Wingate anaerobic test on a cycle ergometer (Velotron; Racermate, Inc., Seattle, WA, USA). Before the initiation of testing, the seat was adjusted to the frame for each participant. The participants were instructed to pedal against zero resistance to become accustomed to the bike and warm-up for

5 minutes at 75–100 revolutions per minute (RPM). After warming up, the participants gradually increased their pedaling speed for 20 seconds in the lead-in phase. Subsequently, they were instructed to reach their maximum pedaling speed in 10 seconds during the speed-up phase, which was followed by the 30-second Wingate phase. During the Wingate phase, the participants continued pedaling as fast as possible against a resistance applied on the flywheel of the cycle ergometer. The resistance was adjusted to allow for 7.5% of the participants' body mass calculated automatically by the Wingate software (Velotron Wingate Software, Version 1.0.2; Racermate, Inc.). After the Wingate phase, the Wingate software reported the peak power (PP), peak revolutions per minute (PRPM), fatigue index (FI), anaerobic capacity (AC), anaerobic power (AP), and total work (TW) values (Table 2). On conclusion of test, the participants continued pedaling with no resistance for 2 minutes to cool-down. During the Wingate phase, strong verbal encouragement was given to encourage participants to achieve their maximal performance.

Aerobic Fitness. Approximately 24 hours after the anaerobic testing, the participants performed maximal oxygen consumption test ($\dot{V}O_{2\max}$) on a motor-driven treadmill (Trackmaster TMX 425C, Newton, KS, USA). Before the initiation of the test, the participants were fitted with a face mask to measure the volume and gas concentrations of inspired and expired air. The expired gases were analyzed using the integrated metabolic measurement system (TrueOne 2400; Parvo Medics, Salt Lake City, UT, USA). Furthermore, the flowmeter was calibrated before each graded exercise test using the 3-L syringe. The participants' heart rates during the aerobic testing were measured through radiotelemetry

TABLE 2. Anaerobic performance.*†

Variable	10-5-HIIT		20-10-HIIT	
	Pre	Post	Pre	Post
Mean power (W)‡	449.5 ± 147.7	507.6 ± 139.1	457.6 ± 109.5	494.3 ± 113.9
Peak power (W)‡	940.7 ± 291.9	1,023.0 ± 306.6	838.8 ± 161.6	886.5 ± 180.6
Minimum power (W)‡	268.9 ± 110.8	301.8 ± 104.7	263.8 ± 65.5	292.4 ± 85.4
Mean speed (rpm)‡	75.7 ± 15.8	85.9 ± 14.8	88.2 ± 14.2	95.6 ± 13.0
Peak speed (rpm)‡	156.1 ± 15.9	169.4 ± 15.5	160.5 ± 15.5	170.3 ± 16.4
Minimum speed (rpm)‡	44.6 ± 13.2	50.4 ± 12.3	52.3 ± 12.1	55.1 ± 13.1
Anaerobic capacity ($W \cdot kg^{-1}$)‡	5.5 ± 1.2	6.3 ± 1.1	6.5 ± 1.0	7.0 ± 1.0
Anaerobic power ($W \cdot kg^{-1}$)‡	11.5 ± 1.2	12.5 ± 1.1	11.8 ± 1.1	12.5 ± 1.2
Fatigue index ($W \cdot s^{-1}$)‡	22.5 ± 7.6	24.2 ± 8.0	19.0 ± 5.1	20.2 ± 5.5
Total work (J)‡	13,482.7 ± 4,432.6	15,225.8 ± 4,172.1	13,727.2 ± 3,284.3	14,827.1 ± 3,412.0

*HIIT = high-intensity interval training.

†Mean ± SD.

‡Significant main effect of time.

TABLE 3. Aerobic performance.*†

Variable	10-5-HIIT		20-10-HIIT	
	Pre	Post	Pre	Post
Time to exhaustion (min:sec)‡	13:38 ± 01:14	14:05 ± 01:05	14:14 ± 01:35	15:00 ± 01:32
$\dot{V}O_{2\max}$ (L·min ⁻¹)‡	3.8 ± 1.3	4.0 ± 1.2	3.6 ± 1.2	3.8 ± 1.2
$\dot{V}O_{2\max}$ (ml·kg ⁻¹ ·min ⁻¹)‡	36.7 ± 7.5	39.1 ± 6.0	39.7 ± 9.3	42.6 ± 9.2
Ventilatory equivalent (L·min ⁻¹)‡	117.6 ± 33.2	130.2 ± 30.2	117.2 ± 35.5	127.0 ± 33.4
Heart rate (b·min ⁻¹)	189.2 ± 12.6	190.4 ± 10.6	192.5 ± 8.1	190.5 ± 5.6
$\dot{V}CO_2$ (L·min ⁻¹)‡	4.5 ± 1.6	4.9 ± 1.5	4.3 ± 1.5	4.6 ± 1.5
Metabolic equivalents (METs)‡	10.5 ± 2.2	11.2 ± 1.7	11.4 ± 2.7	12.2 ± 2.6
Ventilatory efficiency‡	28.2 ± 3.4	29.8 ± 3.5	29.4 ± 2.0	30.2 ± 2.4

*HIIT = high-intensity interval training.

†Mean ± SD.

‡Significant main effect of time.

heart rate monitors (Polar Electro, Inc., Port Washington, NY, USA). After a 3-minute warm-up at 3 mph with 0% grade, the participants completed the Bruce $\dot{V}O_{2\max}$ protocol, which began with the first stage at 1.7 mph and 10% incline for 3 minutes. During the next stages, the speed was increased to 2.5, 3.4, 4.2, 5.0, 5.5, and 6.0 mph, respectively, and the grade was simultaneously elevated by 2% every 3 minutes until volitional exhaustion. The participants' time to exhaustion (TE), absolute $\dot{V}O_{2\max}$ (A- $\dot{V}O_{2\max}$), relative $\dot{V}O_{2\max}$ (R- $\dot{V}O_{2\max}$), ventilatory equivalent (VE), volume of carbon dioxide ($\dot{V}CO_2$), metabolic equivalents (METs), and VE ($\dot{V}CO_2$) were recorded during the test to evaluate the aerobic fitness. Strong verbal encouragement was given to participants throughout the $\dot{V}O_{2\max}$ test to encourage them to perform their maximal effort.

Training Interventions. The training sessions consisted of a 5-minute self-paced jogging to warm-up, a training protocol, and a 5-minute jogging to cool-down. The HIIT protocols involved 6 cycles of 6 functional fitness exercises. As addressed in the previous literature (7,13), the intensity of exercise was determined by "all-out" performance during each bout to reach the intensity zone at approximately 90% of HRmax. The training sessions of both groups consisted of kettle bell snatches, step-up jumps, jumping jacks, goblet squat, burpees, and high knees. The participants of the 10-5-HIIT group performed each training session at a 10 s:5 s work-to-rest ratio followed by a 1-minute recovery after each cycle. Conversely, the participants of the 20-10-HIIT group performed each training session at a 20 s:10 s work-to-rest ratio, and a 2-minute recovery between the cycles. During the exercise sessions, the participants' heart rates were monitored continuously during and after each bout using heart rate monitors (Polar Electro, Inc.) to ensure that the

participants were performing at adequate effort to reach the intensity zone (~90% of HRmax).

Because the primary objective during each bout was to maintain exercise at ~90% of HRmax, to accomplish this goal, we used light kettlebells (men: 20 or 25 lbs and women: 10 or 15 lbs) during both the kettlebell snatch and goblet squat exercises. Furthermore, participants were asked to perform these exercises continuously, and as fast as possible during each bout, and if they were fatigued, the participants were authorized to reduce the load (i.e., use lighter weights) to maintain their goal heart rate (~90% of HRmax). Thus, while participants were able to perform more repetitions in the beginning of exercise session, the goal heart rate was maintained within the target intensity zone, of ~90% of HRmax. The training sessions were performed 3 days a week for a total of 4 weeks.

Statistical Analyses

Before comparing the data, the variables were tested for normality using the Shapiro-Wilk's test. The Mauchly's test was also applied to evaluate the assumption of sphericity for repeated-measures analysis of variance (ANOVA). Statistical significance was selected at $p \leq 0.05$.

A two-way repeated-measures ANOVA with factors group (i.e., 10-5-HIIT and 20-10-HIIT) and time (i.e., pre and post) was applied to analyze the effects of the 2 HIIT protocols on anaerobic and aerobic performance. The main effects were analyzed for the variables that showed no significant interaction effects. As recommended previously (17), the effect size was determined by partial eta-squared (η_p^2) and interpreted as small (0.01), medium (0.06), and large (0.14). The statistical analyses were conducted using IBM SPSS Statistics 23.0 (IBM Corp., Armonk, NY, USA). The alpha (α) level was set at $p \leq 0.05$, and all results were expressed as mean ± SD.

RESULTS

The results obtained from the participants of the 10-5-HIIT and 20-10-HIIT groups indicated no significant difference between the groups for age and height ($p > 0.05$), except for body mass, for which a significant difference in pre-body mass was observed between the 10-5-HIIT and the 20-10-HIIT ($p = 0.04$). Moreover, no significant difference was observed between the groups for any of the pretest variables ($p > 0.05$ in all cases).

Anaerobic Performance

Table 2 summarizes the preintervention and postintervention changes in anaerobic fitness for the 10-5-HIIT and 20-10-HIIT protocols. Overall, similar changes were observed

for all dependent variables in both groups, and consequently, no significant between-group interactions were observed in PP ($F(1, 32) = 2.997, p > 0.05, \eta_p^2 = 0.086$), PRPM ($F(1, 32) = 2.057, p > 0.05, \eta_p^2 = 0.060$), FI ($F(1, 32) = 1.167, p > 0.05, \eta_p^2 = 0.035$), AC ($F(1, 32) = 0.012, p > 0.05, \eta_p^2 = 0.001$), AP ($F(1, 32) = 2.351, p > 0.05, \eta_p^2 = 0.068$), and TW ($F(1, 32) = 3.327, p > 0.05, \eta_p^2 = 0.094$). However, a significant main effect of time was observed for both groups in PP ($F(1, 32) = 16.126, p < 0.05, \eta_p^2 = 0.335$), PRPM ($F(1, 32) = 63.942, p < 0.05, \eta_p^2 = 0.666$), FI ($F(1, 32) = 6.021, p < 0.05, \eta_p^2 = 0.158$), AC ($F(1, 32) = 16.137, p < 0.05, \eta_p^2 = 0.335$), AP ($F(1, 32) = 63.746, p < 0.05, \eta_p^2 = 0.666$), and TW ($F(1, 32) = 43.486, p < 0.05, \eta_p^2 = 0.576$). The significant increases were specifically observed in both 10-5-HIIT and 20-10-HIIT protocols for PP (by $9.2 \pm 6.0\%$ and $5.7 \pm 6.0\%$, respectively), PRPM (by $8.8 \pm 6.0\%$ and $6.2 \pm 5.4\%$, respectively), FI (by $8.8 \pm 14.3\%$ and $7.4 \pm 18.0\%$, respectively), AC (by $14.8 \pm 9.5\%$ and $8.6 \pm 10.0\%$, respectively), AP (by $9.0 \pm 6.2\%$ and $6.2 \pm 5.3\%$, respectively), and TW (by $15.1 \pm 10.0\%$ and $8.5 \pm 11.0\%$, respectively).

Aerobic Performance

Table 3 summarizes the prere-sult and postresult of the graded exercise tests. Statistical analysis indicated no significant time $\times$ group interactions in TE ($F(1, 32) = 1.996, p > 0.05, \eta_p^2 = 0.059$), A- $\dot{V}O_{2\max}$ ($F(1, 32) = 0.046, p > 0.05, \eta_p^2 = 0.001$), R- $\dot{V}O_{2\max}$ ($F(1, 32) = 0.091, p > 0.05, \eta_p^2 = 0.003$), VE ($F(1, 32) = 0.288, p > 0.05, \eta_p^2 = 0.009$), $\dot{V}CO_2$ ($F(1, 32) = 0.001, p > 0.05, \eta_p^2 = 0.001$), METs ($F(1, 32) = 0.097, p > 0.05, \eta_p^2 = 0.003$), and VE: $\dot{V}CO_2$ ($F(1, 32) = 2.045, p > 0.05, \eta_p^2 = 0.060$). However, a significant main effect of time was observed in both 10-5-HIIT and 20-10-HIIT for TE ($F(1, 32) = 27.670, p < 0.05, \eta_p^2 = 0.464$), A- $\dot{V}O_{2\max}$ ($F(1, 32) = 12.600, p < 0.05, \eta_p^2 = 0.283$), R- $\dot{V}O_{2\max}$ ($F(1, 32) = 11.475, p < 0.05, \eta_p^2 = 0.003$), VE ($F(1, 32) = 18.538, p < 0.05, \eta_p^2 = 0.367$), $\dot{V}CO_2$ ($F(1, 32) = 10.020$,

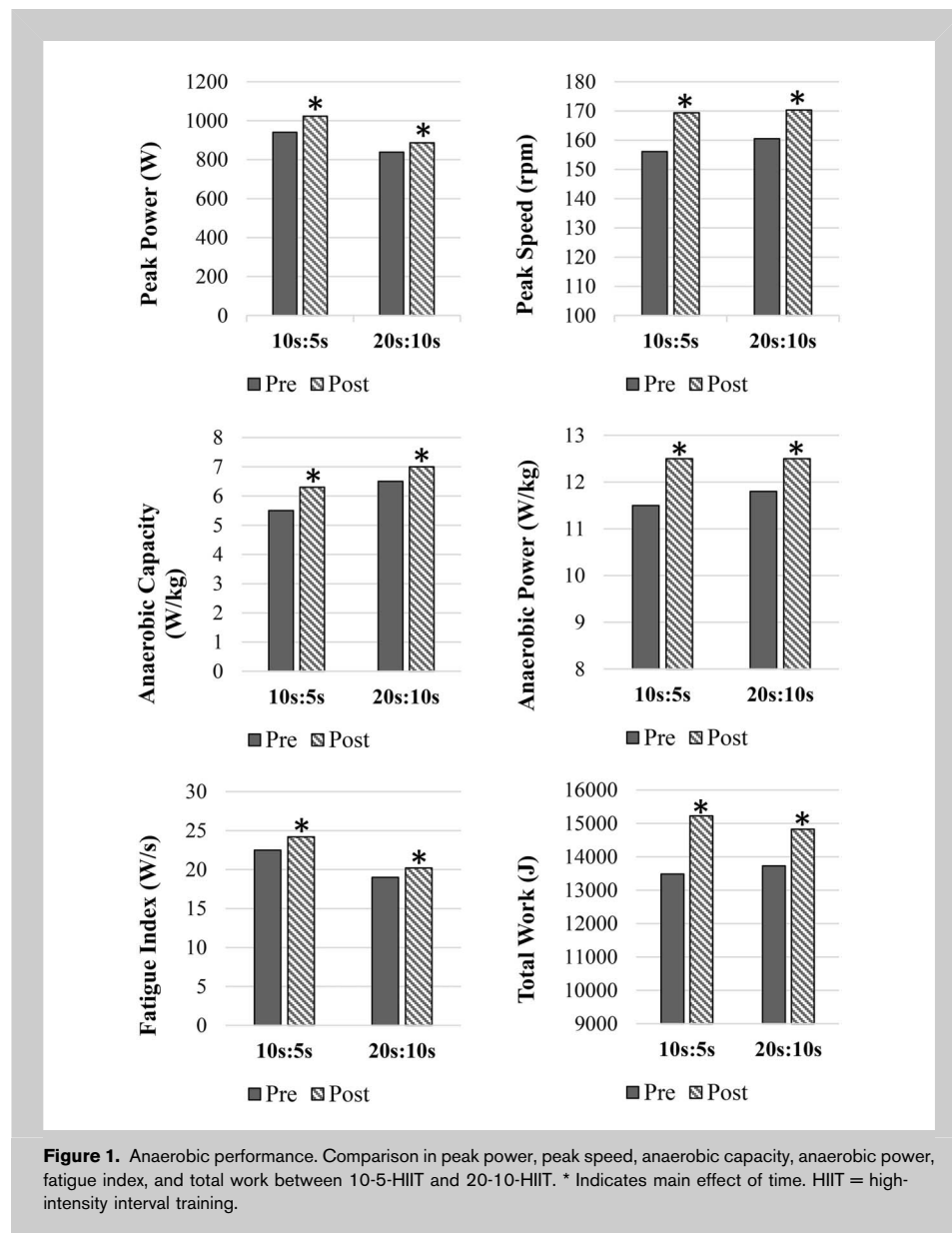


Figure 1. Anaerobic performance. Comparison in peak power, peak speed, anaerobic capacity, anaerobic power, fatigue index, and total work between 10-5-HIIT and 20-10-HIIT. * Indicates main effect of time. HIIT = high-intensity interval training.

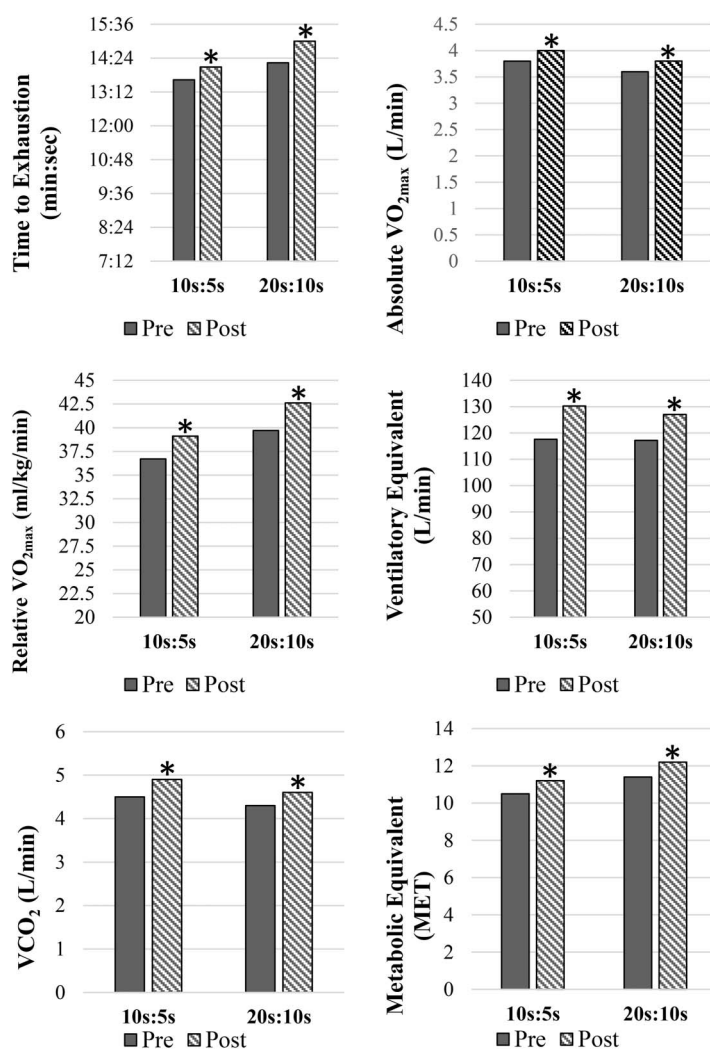


Figure 2. Aerobic performance. Comparison in time to exhaustion, absolute $\dot{V}O_{2max}$, relative $\dot{V}O_{2max}$, ventilatory equivalent, $\dot{V}CO_2$, and metabolic equivalent. METs = metabolic equivalents. * Indicates main effect of time.

$p < 0.05$, $\eta_p^2 = 0.238$), METs ($F(1, 32) = 11.561$, $p < 0.05$, $\eta_p^2 = 0.265$), and VE: $\dot{V}CO_2$ ($F(1, 32) = 18.972$, $p < 0.05$, $\eta_p^2 = 0.372$). Both groups significantly increased TE (by $4.3 \pm 4.0\%$ and $5.5 \pm 4.5\%$, respectively), A- $\dot{V}O_{2max}$ (by $9.4 \pm 16.6\%$ and $8.9 \pm 11.2\%$, respectively), R- $\dot{V}O_{2max}$ (by $8.5 \pm 16.2\%$ and $8.2 \pm 11.4\%$, respectively), VE (by $12.6 \pm 14.2\%$ and $9.8 \pm 12.9\%$, respectively), $\dot{V}CO_2$ (by $11.4 \pm 19.0\%$ and $10.0 \pm 12.5\%$ respectively), METs (by $8.6 \pm 16.4\%$ and $8.3 \pm 11.6\%$, respectively), and VE: $\dot{V}CO_2$ (by $5.7 \pm 6.6\%$ and $2.7 \pm 4.4\%$, respectively).

DISCUSSION

To our knowledge, this is the first study that compares the effects of 2 HIIT configurations, a 10-5 vs. a 20-10 second work-to-rest ratio, on anaerobic and aerobic performance.

Several studies in literature have reported the efficiency of HIIT in improving physical fitness, as an alternative to MICT (10–12). Considering as the primary benefit, HIIT induces similar or even greater physiological adaptations in a shorter time compared with traditional endurance training (11). Therefore, HIIT is proven a time-efficient form of training (9), which can mitigate the most significant barrier to physical activity that is “lack of time” (20). In the past, HIIT protocols were typically performed through running- or ergometer-based exercise, and most previous HIIT studies primarily implemented the Wingate cycle ergometer test as the protocol (9,10,12,22,23,28), which may not be easily accessible for general population. In this study, however, functional fitness exercises were performed during training interventions, including combinations of various total body exercises. Previously, functional fitness training was recommended for developing and maintaining motor skills, cardiorespiratory, neuromotor, and muscular fitness (8). Although the physiological adaptations of HIIT depend on several factors, the intensity

and duration of HIIT intervals is known to have the most impact on physiological adaptations (1). Although a variety of fixed work-to-rest ratios (i.e., 2:1, 1:1, 1:2), or fixed percentage of HRmax are usually used during HIIT protocol, a 2:1 work-to-rest ratio induces higher accumulated oxygen deficit than other protocols. Previous literature has reported that the most effective and shortest duration 2:1 ratio uses 20 s:10 s work-to-rest ratio using a cycle ergometer (15,23). This is the first study to develop and test a shorter HIIT interval performed at a 10 s:5 s work-to-rest ratio, with the goal of determining whether shorter intervals can elicit similar adaptations. Both the HIIT bouts and the rest periods of the 10-5-HIIT (10 s and 5 s, respectively) were half as short as the HIIT bouts and the rest periods of the 20-10-HIIT (20 s and 10 s, respectively). Because of the shorter bout duration, the 10-5-HIIT group was able to maintain a higher

intensity during a 10-s bout compared with the 20-10-HIIT, even with a shorter recovery after each bout.

Despite the differences in their exercise volumes and durations, both HIIT protocols in conjunction with functional fitness training induced similar increases in anaerobic fitness (Table 2). These results are in agreement with previous reports that used the Wingate cycle ergometer assessment protocol (9,10,12,22,23). The load of the outside resistance (i.e., kettlebells) was minimal, and all participants were able to perform the exercises continuously as fast as possible to reach and maintain the intensity zone (i.e., ~90% of HRmax). Specifically, performing bouts that use fast total body exercises near the maximal effort (i.e., ~90% HRmax) improved the anaerobic performance (Figure 1), similar to previous studies. Our results indicate that both training groups significantly enhanced PP and PRPM after 4 weeks of training. Similarly, the FI, defined as the rate of decline in speed and power during performance (19), increased in both groups after training intervention. Although both PP and PRPM significantly increased from pre-testing to post-testing, the minimum RPM did not significantly change, which resulted in higher drop-off in speed during the 30 s Wingate phase in post-testing (19). Therefore, our results indicate that the participants of both groups improved their ability to produce power, which relies primarily on the anaerobic glycolytic energy pathways and may further explain the resulting increases in AC, AP, and TW (Figure 1). Furthermore, the participants of the 10-5-HIIT group significantly improved their anaerobic performance similar to the participants of the 20-10-HIIT group, but with a shorter time commitment to exercise performance.

Several studies have shown that HIIT leads to greater physical adaptations compared with traditional MICT (5,6,29). Specifically, HIIT has been suggested to induce a more significant increase in aerobic capacity compared with MICT (3,18,30). Moreover, HIIT has been reported to significantly increase peak oxygen uptake and $\dot{V}O_{2\max}$ in recreationally active and sedentary individuals, compared with MICT (26). In this study, the participants of both 10-5-HIIT and 20-10-HIIT protocols significantly improved their aerobic capacity (Table 3). Although no significant differences observed between the groups, both HIIT protocols indicated similar increases in TE, A- $\dot{V}O_{2\max}$, R- $\dot{V}O_{2\max}$, VE, $\dot{V}CO_2$, METs, and VE: $\dot{V}CO_2$ after the training interventions (Figure 2). Our results indicated that the participants of the 10-5-HIIT group were able to significantly enhance their aerobic fitness, similar to the 20-10-HIIT protocol but over a shorter duration of time (Figure 2). Similar to the anaerobic performance, the 10-5-HIIT interval at 10 s:5 s work-to-rest ratio is proposed as the shortest and most efficient interval for improving aerobic capacity.

There are several limitations of this study. A between-subject design was used to examine the effects of 10-5-HIIT and 20-10-HIIT on anaerobic and aerobic performance. Sex was not included as a factor, which may have contributed to

increased variability in our data. Future studies are needed to examine whether there is a differential effect of these protocols on performance for men vs. women. In addition, the participants in this study would be best categorized as low fitness; therefore, caution is warranted when extrapolating these results to participants with higher fitness levels.

The findings of this study indicate that HIIT may serve as a time-efficient type of exercise to improve health-related physical fitness. Indeed, our results suggest that performing functional fitness training at high effort levels (~90% of HRmax) using HIIT protocols seems to be beneficial for anaerobic and aerobic performance. After 4 weeks of 10-5-HIIT and 20-10-HIIT, both groups similarly improved anaerobic and aerobic performance. The results confirm that both protocols can induce similar performance adaptations. However, our results suggest that performing HIIT at a 10 s:5 s work-to-rest ratio (i.e., the 10-5-HIIT) can improve anaerobic and aerobic fitness with a shorter time commitment. Therefore, this study suggests that performing HIIT at a 10 s:5 s work-to-rest ratio is the shortest and most efficient protocol for improving anaerobic and aerobic performance in participants with low-fitness levels. To our knowledge, this is the first study that uses a 10 s:5 s work-to-rest ratio performing functional fitness exercises during HIIT, which may provide new perspectives for future studies.

PRACTICAL APPLICATIONS

Compared with high-volume continuous training, HIIT is proven effective in inducing similar physiological adaptations with a shorter time commitment. "Lack of time" has had the strongest association with the lack of physical activity in developed countries, which has been identified as the main barrier to not meeting the recommended exercise levels and durations. Hence, HIIT has become an alternative for traditional high-volume endurance training. The results of this study confirm that performing HIIT at 10 s:5 s work-to-rest ratio offers a shorter and equally efficient interval to enhance anaerobic and aerobic performance compared with other protocols. Moreover, using functional fitness training at ~90% HRmax during HIIT seems to be beneficial to improve anaerobic and aerobic performance, especially for those who have limited/no access to fitness equipment and facilities.

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